Reg. No. | | | | | | | | | | [12CS-IS64/13CS-IS64]

SIXTH SEMESTER B.E/B.TECH EXAMINATION, JUNE – 2016

PARALLEL COMPUTING SYSTEMS

Time: 3 Hours **Maximum Marks:70**

Instructions: **1.** Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	**a.**	Explain Hybrid memory architecture with a neat diagram.	**08**
	b.	Compare and contrast SIMD and MISD architectures.	**06**
2.	**a.**	Discuss on different implementations of thread model.	**07**
	b.	Considering a data parallel programming model with 4 tasks, demonstrate how you would add 10 to every array element in an array which contains 100 elements.	**07**
3.	**a.**	Explain the working of a fully automatic and programmer directed parallelizing compiler.	**06**
	b.	Describe any four important factors to consider when designing program's inter-task communications.	**08**
4.	**a.**	Explain loop dependent and loop independent data dependence with an example.	**07**
	b.	State and explain Amdahl's Law with relevant examples.	**07**
5.	**a.**	Discuss about the future applications that are driving for more speed or parallelism.	**04**
	b.	Explain the architecture of a CUDA capable GPU with a neat diagram.	**10**
6.	**a.**	With a neat diagram, explain the working of programmable graphics pipeline with a separate vertex processor and fragment processor.	**10**
	b.	Write a note on scalable GPUs.	**04**
7.	**a.**	Explain platform model with a relevant diagram.	**06**
	b.	What are the types of commands that OpenCL supports? Explain the models in which these commands can be executed.	**08**
8.	**a.**	Mention the steps performed within main () to execute an OpenCL program.	**07**
	b.	How to create and build a program object?	**07**